

AI2002 – Assignment 1 Evaluation Rubric

Section: _____

Student 1 Name: _____ Roll No: _____

Student 2 Name: _____ Roll No: _____

Student 3 Name: _____ Roll No: _____

Part A: Project Proposal Document Evaluation (50 Marks)

1. Problem Statement & Domain Characterization (15 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Problem articulation & real-world relevance (6)	6 – Problem clearly defined, realistic, meaningful application of AI	4 – Clear but slightly generic or less impactful	2 – Somewhat vague or partially defined	0 – Unclear or unrealistic	_____
AI problem classification (5)	5 – Correctly classified with strong justification (search, CSP, optimization etc.)	3 – Correct classification but weak justification	2 – Partial understanding	0 – Incorrect classification	_____
Computational complexity discussion (4)	4 – Clearly explains state space, branching factor, complexity	3 – Mentions complexity but lacks depth	1 – Very limited discussion	0 – Not discussed	_____

2. Objectives & Scope (10 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Objectives clarity (4)	4 – Clear, measurable and well-defined objectives	3 – Mostly clear objectives	2 – Vague objectives	0 – Missing	_____
Scope definition (3)	3 – Clear constraints and assumptions	2 – Some constraints defined	1 – Minimal scope definition	0 – No scope defined	_____

Expected deliverables (3)	3 – Clear expected outcomes and evaluation metrics	2 – Deliverables defined but basic	1 – Deliverables unclear	0 – Missing	_____
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3. Algorithmic Framework & Technique Justification (15 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Algorithm selection (5)	5 – Highly appropriate algorithm for problem	3 – Reasonably appropriate	2 – Weak match	0 – Irrelevant algorithm	_____
Theoretical justification (5)	5 – Strong explanation (optimality, completeness, complexity etc.)	3 – Partial explanation	2 – Limited explanation	0 – None	_____
System architecture / workflow diagram (5)	5 – Clear architecture diagram and explanation	3 – Diagram present but basic	2 – Incomplete design	0 – Missing	_____

4. User Interaction / Interface Design (5 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Interface design explanation (3)	3 – Clear explanation of user interaction and visualization	2 – Some explanation	1 – Minimal explanation	0 – Missing	_____
Framework/tool selection (2)	2 – Properly justified framework choice	1 – Framework mentioned but not justified	1 – Weak reasoning	0 – Not mentioned	_____

5. Overall Document Quality (5 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Structure, formatting, language (5)	5 – Professional formatting, logical flow, minimal errors	3 – Minor formatting issues	2 – Several writing issues	0 – Poorly structured	_____

Part B: Search Algorithm Implementation & Demo (50 Marks)

1. Algorithm Implementation Correctness (15 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Correct implementation of search algorithm (10)	10 – Algorithm implemented correctly and works for all test cases	7 – Mostly correct with minor issues	4 – Partially correct	0 – Incorrect	_____
Domain adaptation (5)	5 – Algorithm clearly integrated with project problem	3 – Some integration	2 – Weak connection	0 – Generic algorithm only	_____

2. Heuristic Function Design (10 Marks) (for informed search)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Heuristic design (5)	5 – Domain-specific, well thought-out	3 – Reasonable heuristic	2 – Basic heuristic	0 – No heuristic	_____
Justification & properties (5)	5 – Admissibility/consistency explained	3 – Partial justification	2 – Weak explanation	0 – None	_____

(If uninformed search, evaluate instead on problem modeling and state representation.)

3. Code Quality & Modularity (10 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Code organization (4)	4 – Modular, well-structured	3 – Mostly organized	2 – Some structure	0 – Poor structure	_____
Comments & readability (3)	3 – Well commented and easy to follow	2 – Some comments	1 – Few comments	0 – No comments	_____
Error handling & validation (3)	3 – Proper validation and handling	2 – Partial validation	1 – Minimal validation	0 – None	_____

4. Demonstration & Functionality (7 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Working demo (4)	4 – Fully functional and stable	3 – Mostly working	2 – Some bugs	0 – Not working	_____

Input/output clarity (2)	2 – Clear input handling and meaningful outputs	1 – Reasonable outputs	1 – Outputs unclear	0 – Poor output	_____
Interface/visualization (1)	1 – GUI or well-designed console interface	1 – Basic interface	0.5 – Minimal effort	0 – Hard to use	_____

5. Conceptual Understanding During Demo (5 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Student explanation of algorithm & design (5)	5 – Clear understanding, answers questions confidently	3 – Mostly understands	2 – Limited understanding	0 – Cannot explain	_____

6. README Evaluation (3 Marks)

Criteria	Excellent	Good	Average	Poor	Marks Awarded
Execution instructions (1)	1 – Clear steps to run the project	0.5 – Some steps missing	0.5 – Unclear instructions	0 – Missing	_____
Dependencies listed (1)	1 – All required libraries/tools listed	0.5 – Partial list	0.5 – Minimal info	0 – Not mentioned	_____
Sample input/output + overview (1)	1 – Includes example run and brief technical overview	0.5 – Partial example	0.5 – Weak explanation	0 – Missing	_____